Nathan Nguyen

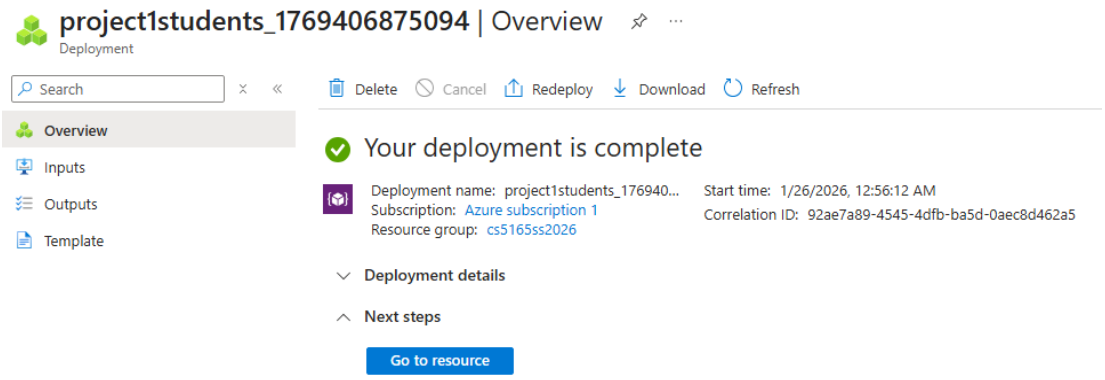
Azure Data Pipeline Project Submission

1. **Set up an Azure Storage account and upload the student.csv file.** (Screenshots of each major step (Azure portal setups, Storage account/Container creation, File Upload results, etc.): **Points: 2**

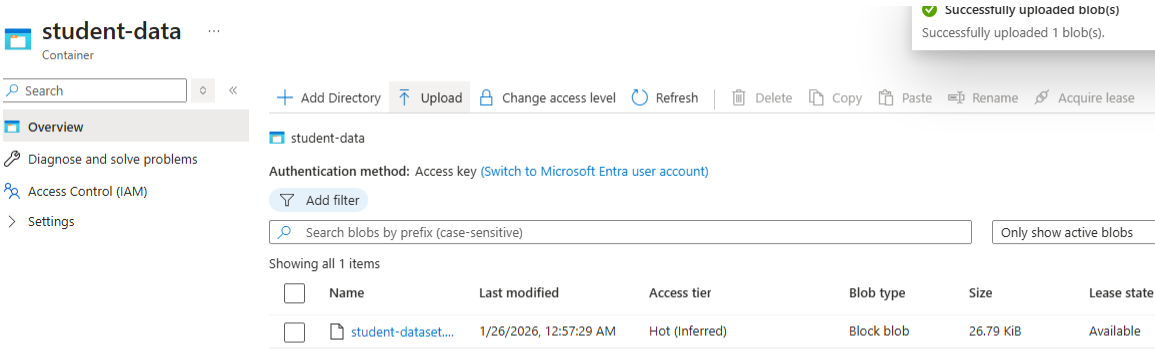
2. **Set up an Azure SQL Database and create the Students table.**(Screenshots of each major step (SQL Server setups, SQL Database creation, Firewall rules, Table creation, etc.): **Points: 2**
3. **Build a Data Pipeline in Azure Data Factory** (Provide screenshots of each major steps): **Points: 2**

Photos of key Steps for 1-3 (labeled based on actual step in guide):

1.1



1.2



2.2

Dashboard › Azure SQL | SQL databases

Create SQL Database

Submitting deployment...

Submitting the deployment template for resource group 'cs5165ss2026'.

Basics | Review + create

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see Azure Marketplace Terms.

Basics

Subscription	Azure subscription 1
Resource Group	cs5165ss2026
Region	East US 2
Database name	students
Server	(new) cs5165-studentsdb
Authentication method	SQL and Microsoft Entra authentication
Microsoft Entra Admin	nathanpn2003@gmail.com
Server admin login	nathanpng
Compute + storage	GeneralPurpose - Serverless - Standard-series (Gen5), 2 vCores, 32 GB Storage, zone redundant disabled
Backup storage redundancy	Locally-redundant backup storage
Overage billing	Disabled

Preconfigured details

Cost summary

Compute	$0.00
vCore second [1]	$0.00
Storage	$0.00
Cost per GB	$0.00
Max storage selected (in GB)	32
Overage billing	Disabled

There will be no charges for usage within the free limits. The database will be paused automatically when the free limits are reached.

Estimated total: Free [1]

Prices reflect an estimate only. View Azure pricing calculator. Final charges will appear in your local currency in cost analysis and billing views.

[1] Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. Learn more about serverless billing

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2.3

studentws (nathanpn2003@gmail.com

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

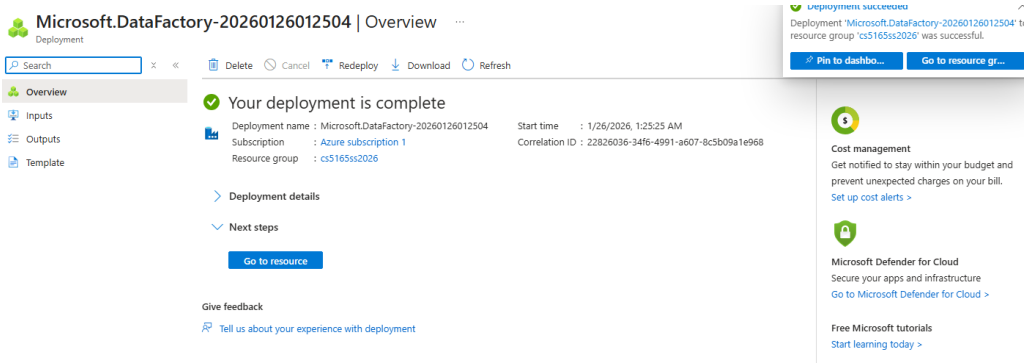
- Tables
  - dbo.Students
- Views
- Stored Procedures

Query 1

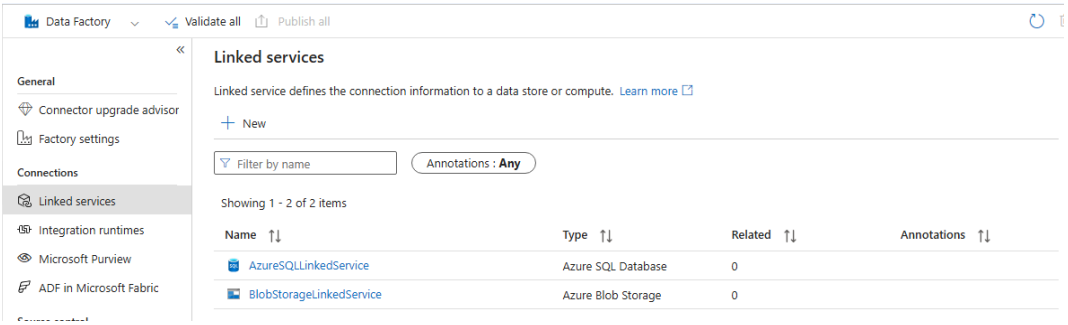
Run | Cancel query | Save query | Export data as | Show only Editor

```sql
CREATE TABLE Students (
    StudentID INT NOT NULL PRIMARY KEY,
    Name NVARCHAR(100),
    Country NVARCHAR(100),
    City NVARCHAR(100),
    Latitude FLOAT,
    Longitude FLOAT,
    Gender NVARCHAR(10),
    EthnicGroup NVARCHAR(50),
    Age INT,
    EnglishGrade FLOAT,
    MathGrade FLOAT,
    SciencesGrade FLOAT,
    LanguageGrade INT,
    PortfolioRating INT,
    CoverletterRating INT,
    RefletterRating INT
);
```

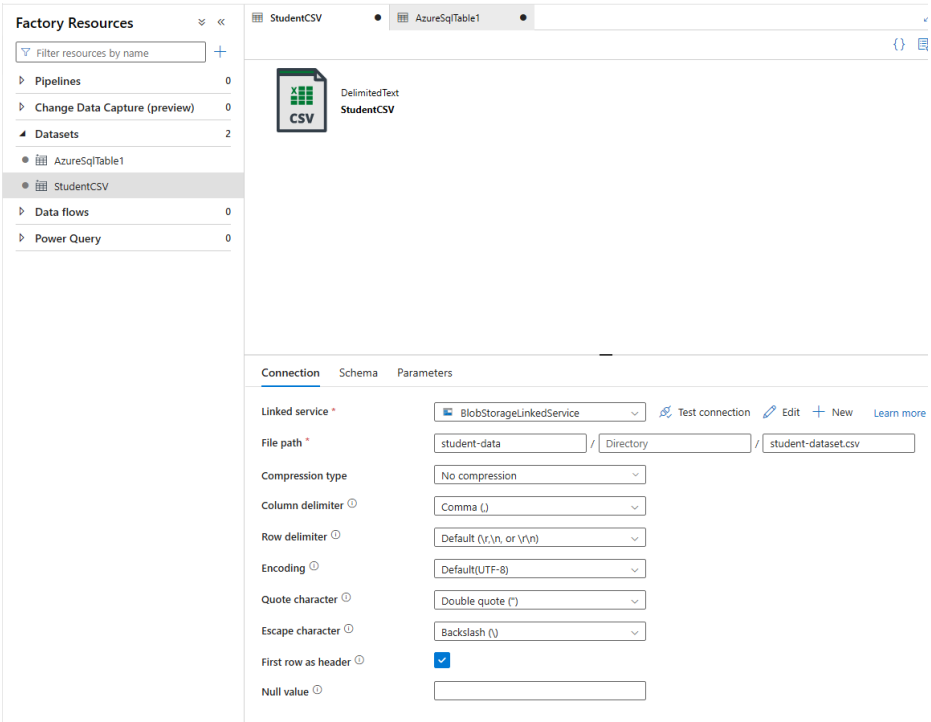
3.1


Microsoft.DataFactory-20260126012504 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
Your deployment is complete
Deployment name : Microsoft.DataFactory-20260126012504
Subscription : Azure subscription 1
Resource group : cs5165ss2026
Start time : 1/26/2026, 1:25:25 AM
Correlation ID : 22826036-34f6-4991-a607-8c5b09a1e968
Deployment details
Next steps
Go to resource
Give feedback
Tell us about your experience with deployment
Pin to dashbo...
Go to resource gr...
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
Start learning today >

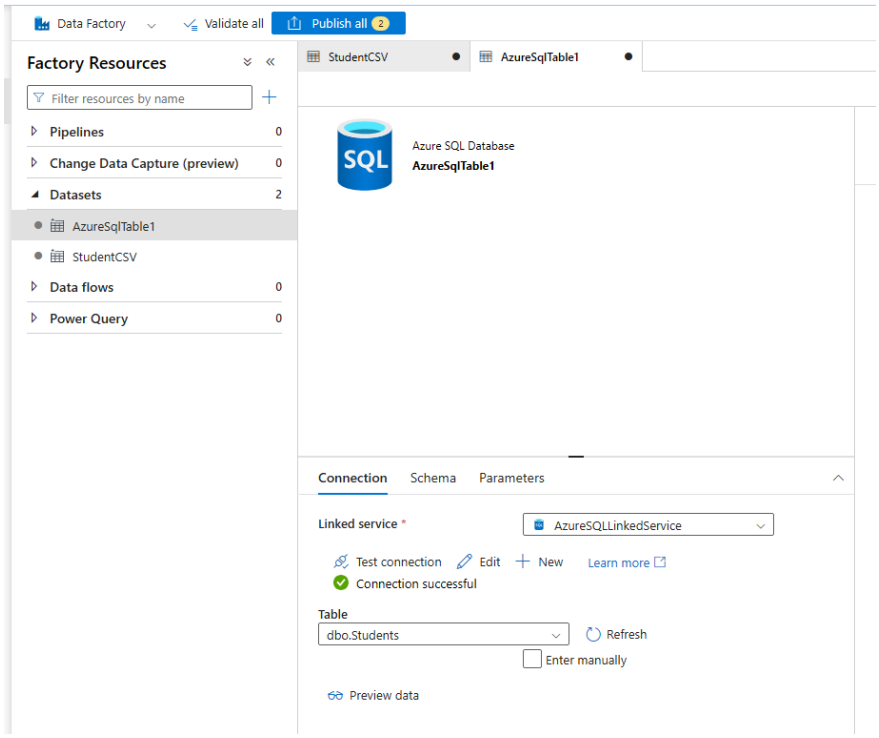
4.2


Data Factory
Validate all
Publish all
General
Connector upgrade advisor
Factory settings
Connections
Linked services
Integration runtimes
Microsoft Purview
ADF in Microsoft Fabric
Linked services
Linked service defines the connection information to a data store or compute. Learn more
New
Filter by name
Annotations : Any
Showing 1 - 2 of 2 items
Name
Type
Related
Annotations
AzureSQLLinkedService
Azure SQL Database
0
BlobStorageLinkedService
Azure Blob Storage
0

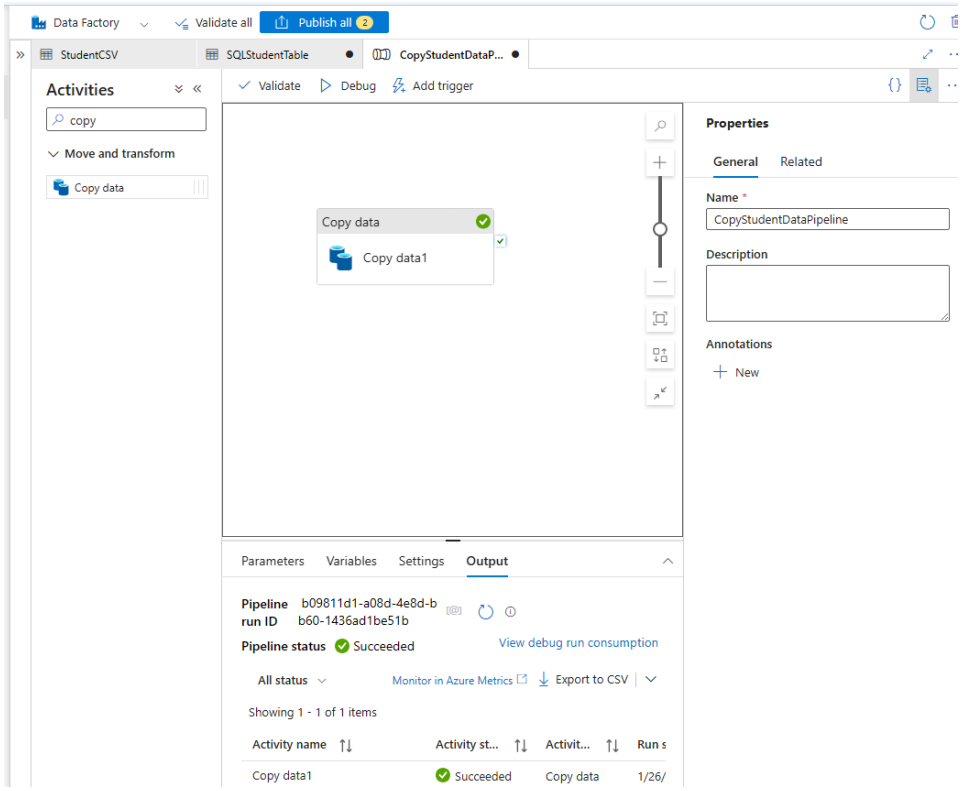
5.1


Factory Resources
Filter resources by name
Pipelines 0
Change Data Capture (preview) 0
Datasets 2
AzureSqlTable1
StudentCSV
Data flows 0
Power Query 0
StudentCSV
AzureSqlTable1
CSV
DelimitedText
StudentCSV
Connection
Schema
Parameters
Linked service *
BlobStorageLinkedService
Test connection
Edit
New
Learn more
File path *
student-data
Directory
student-dataset.csv
Compression type
No compression
Column delimiter
Comma (,)
Row delimiter
Default (\r,\n, or \r\n)
Encoding
Default(UTF-8)
Quote character
Double quote (")
Escape character
Backslash (\)
First row as header
Null value

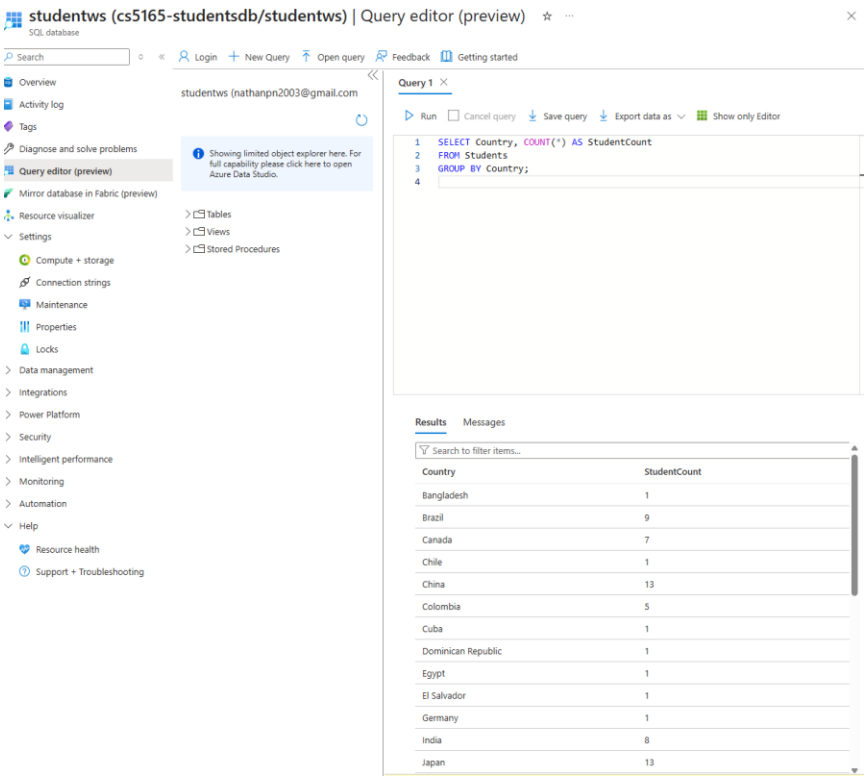
5.2


Data Factory
Validate all
Publish all 2
Factory Resources
Filter resources by name
Pipelines 0
Change Data Capture (preview) 0
Datasets 2
AzureSqlTable1
StudentCSV
Data flows 0
Power Query 0
StudentCSV
AzureSqlTable1
Azure SQL Database
AzureSqlTable1
Connection
Schema
Parameters
Linked service *
AzureSQLLinkedService
Test connection
Edit
New
Learn more
Connection successful
Table
dbo.Students
Refresh
Enter manually
Preview data

6.4


Data Factory
Validate all
Publish all 2
StudentCSV
SQLStudentTable
CopyStudentDataP...
Activities
copy
Move and transform
Copy data
Validate
Debug
Add trigger
Copy data
Copy data1
Properties
General
Related
Name *
CopyStudentDataPipeline
Description
Annotations
New
Parameters
Variables
Settings
Output
Pipeline run ID
b09811d1-a08d-4e8d-b b60-1436ad1be51b
Pipeline status
Succeeded
View debug run consumption
All status
Monitor in Azure Metrics
Export to CSV
Showing 1 - 1 of 1 items
Activity name
Activity st...
Activit...
Run s
Copy data1
Succeeded
Copy data
1/26/

6.5



4. **Provide the Connection information in a text format for me to remotely connect to your Azure database from** SQL Server Management Studio (SSMS) in my machine and run the Query: **Points: 2**

    - **Authentication**: SQL Server Authentication
    - **Server name**: cs5165-studentsdb.database.windows.net
    - Database name: studentws
    - **Login**: nathanpng
    - **Password**: January2026
    - Ensure the firewall settings and credentials are correct.

        1. Firewall settings:
           - Public network access enabled
           - Allow Azure services and resources to access this server
           - External firewall rules configured for SSMS access\

5. **High-level pricing estimate** for the Azure Services used in the project per month: **Point: 1**

- *** Cost needs to be broken down by Azure Services, with a short explanation on how you are projecting the final cost

Azure Services used & Estimated Cost:

- **Azure SQL Database**
  The project uses an Azure SQL Database deployed on the Free or Basic DTU-based tier, which is suitable for small datasets and low query volume. Since the database stores a single table populated from a CSV file and is accessed infrequently for testing and demonstration, the estimated cost is **$0–$5 per month**.
- **Azure Data Factory**
  Azure Data Factory is used to run a single pipeline containing a Copy Data activity to ingest the CSV file from Azure Blob Storage into Azure SQL Database. Costs are based on the number of pipeline and activity runs. Because the pipeline was executed only a small number of times during development and testing, the estimated monthly cost is **less than $2**.
- **Azure Blob Storage**
  Azure Blob Storage (Locally Redundant Storage) is used to store a single CSV file containing student data. The file size is very small, resulting in negligible storage and transaction costs. The estimated monthly cost is **less than $1**.
- **Azure Functions (Extra Credit)**
  An Azure Function running on the Consumption plan is used to query the Azure SQL Database and expose the results as a REST API. Since the function is invoked only a limited number of times and remains well within the free execution grant, the estimated cost is **$0 per month**.
- **Azure Static Web Apps (Extra Credit)**
  The frontend is hosted using Azure Static Web Apps on the Free tier. This tier supports static content hosting at no cost for small projects. The estimated monthly cost is **$0**.
- **Estimated Total Monthly Cost**
  The overall estimated cost of the project is **approximately $0–$8 per month**, depending on database tier usage and pipeline executions.

6. **Provide a short explanation of your project, including objectives, tools used, implementation steps, and "Student Count by Country" results**: **Points: 1**
   - The objective of this project was to design and implement an end-to-end data pipeline using Microsoft Azure that ingests student data from a CSV file,

stores it in a relational database, and enables analysis through both SQL queries and a web interface.

The project utilizes several Azure services. Azure Blob Storage is used to store the source CSV file containing student information. Azure Data Factory is responsible for orchestrating the data movement by copying the data from Blob Storage into an Azure SQL Database. The student data is stored in a Students table, which allows structured querying and aggregation.

After the data was loaded, a SQL query was executed to calculate the number of students grouped by country. For the extra credit component, an Azure Function was implemented to query the database and expose the results as a REST API. An Azure Static Web App consumes this API and dynamically displays the results in a web-based table.

The final "Student Count by Country" results show that the United States of America has the highest number of students (193), followed by Mexico (24), China (13), and Japan (13), with several other countries represented by smaller counts. This confirms that the data pipeline successfully ingested, stored, and processed the dataset end-to-end.

- Student Count by Country Results:
  United States of America: 193
  Mexico: 24
  China: 13
  Japan: 13
  Brazil: 9
  India: 8
  Canada: 7
  Russian Federation: 6
  Colombia: 5
  Pakistan: 3
  Korea (Republic of): 3
  Turkey: 2
  Spain: 2
  United Kingdom: 2
  Thailand: 1
  Tunisia: 1
  Ukraine: 1
  Peru: 1
  Philippines: 1

Poland: 1
Morocco: 1
Myanmar: 1
Netherlands: 1
Nicaragua: 1
Cuba: 1
Dominican Republic: 1
Egypt: 1
El Salvador: 1
Germany: 1
Chile: 1
Bangladesh: 1

7. Extra Credit: **(Points: 1)** Some students have expressed interest in exploring cutting-edge cloud technologies, and Azure Functions is one of them. This serverless solution is widely used in the industry for building event-driven applications with minimal code and infrastructure, making it ideal for web APIs, data processing, and IoT integration. This extra credit assignment will guide you through creating a simple web app to visualize student data by country.

   - Provide the URL to the Static Web App, ensure it is accessible, and confirm that it correctly fetches and displays the "Student Count by Country": **Points: 1**

     1. https://proud-pond-03afc530f.1.azurestaticapps.net/

        NOTE: It might take 60-90 seconds before the table populates, seems like one of my runners sometimes drains randomly, not sure why

Student Count by Country

Country	Student Count
United States of America	193
Mexico	24
China	13
Japan	13
Brazil	9
India	8
Canada	7
Russian Federation	6
Colombia	5
Pakistan	3
Korea (Republic of)	3
Turkey	2
Spain	2
United Kingdom	2
Thailand	1
Tunisia	1
Ukraine	1
Peru	1
Philippines	1
Poland	1
Morocco	1
Myanmar	1
Netherlands	1
Nicaragua	1
Cuba	1
Dominican Republic	1
Egypt	1
El Salvador	1
Germany	1
Chile	1